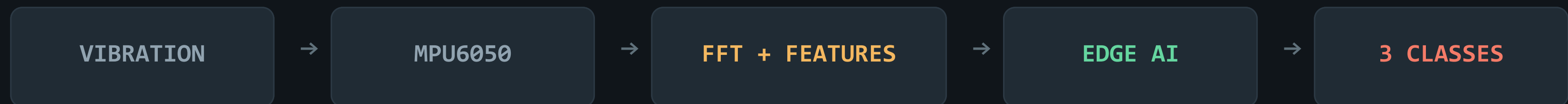
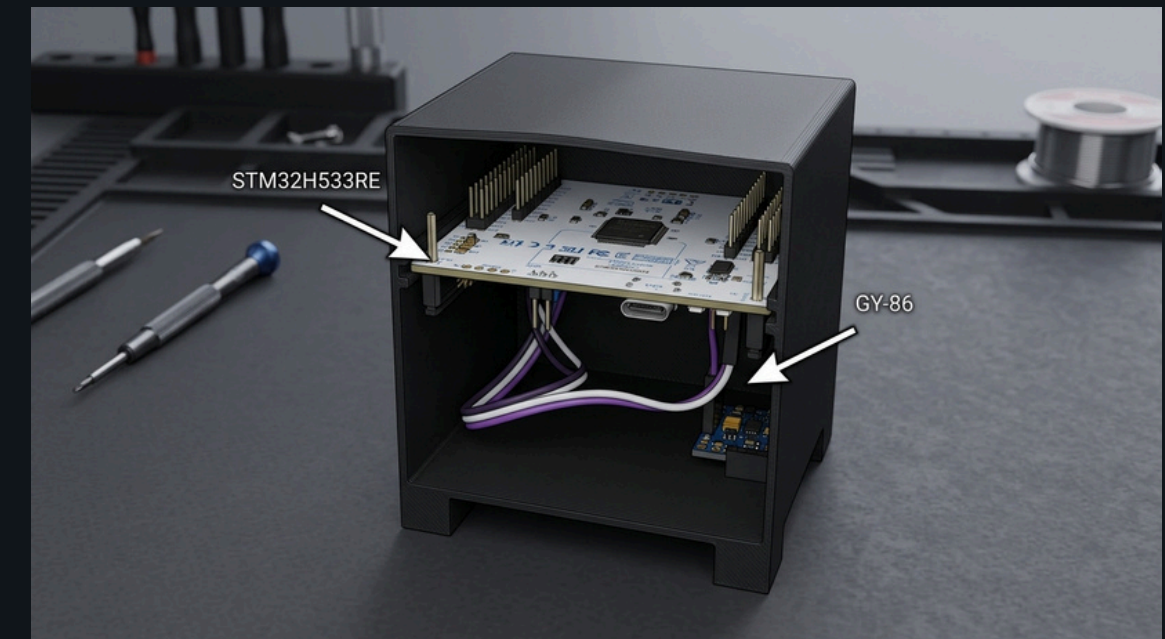


VibeFlow

Real-time vibration classification on STM32

Measure → transform → classify → visualize



From raw vibration to an actionable class

The system is designed as a complete sensing-to-decision pipeline.

THE CORE QUESTION

“What vibration state is the object experiencing right now?”

VibeFlow answers this from accelerometer data instead of relying on a host computer for the inference step.

- EDGE INFERENCE
- REAL-TIME
- 3 CLASSES

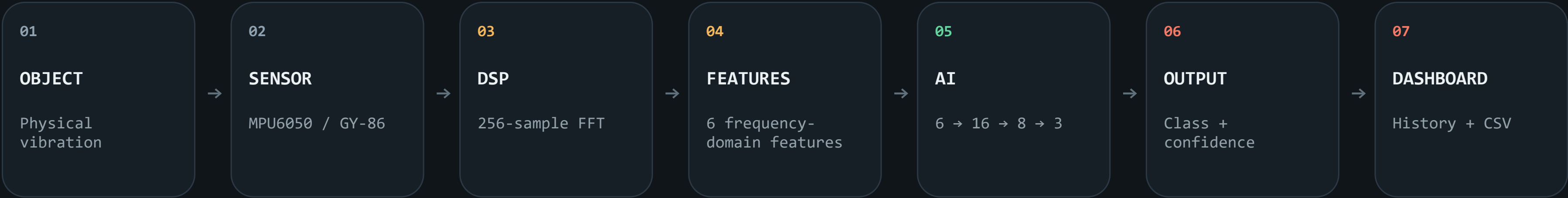
WHAT THE USER SEES

Stationary	No meaningful vibration	01
Low vibration	Light vibration	02
High vibration	Strong vibration	03

The class is accompanied by a confidence value and stored in the detection history.

One signal, one embedded decision

VibeFlow keeps acquisition, signal processing, inference, and interaction connected.



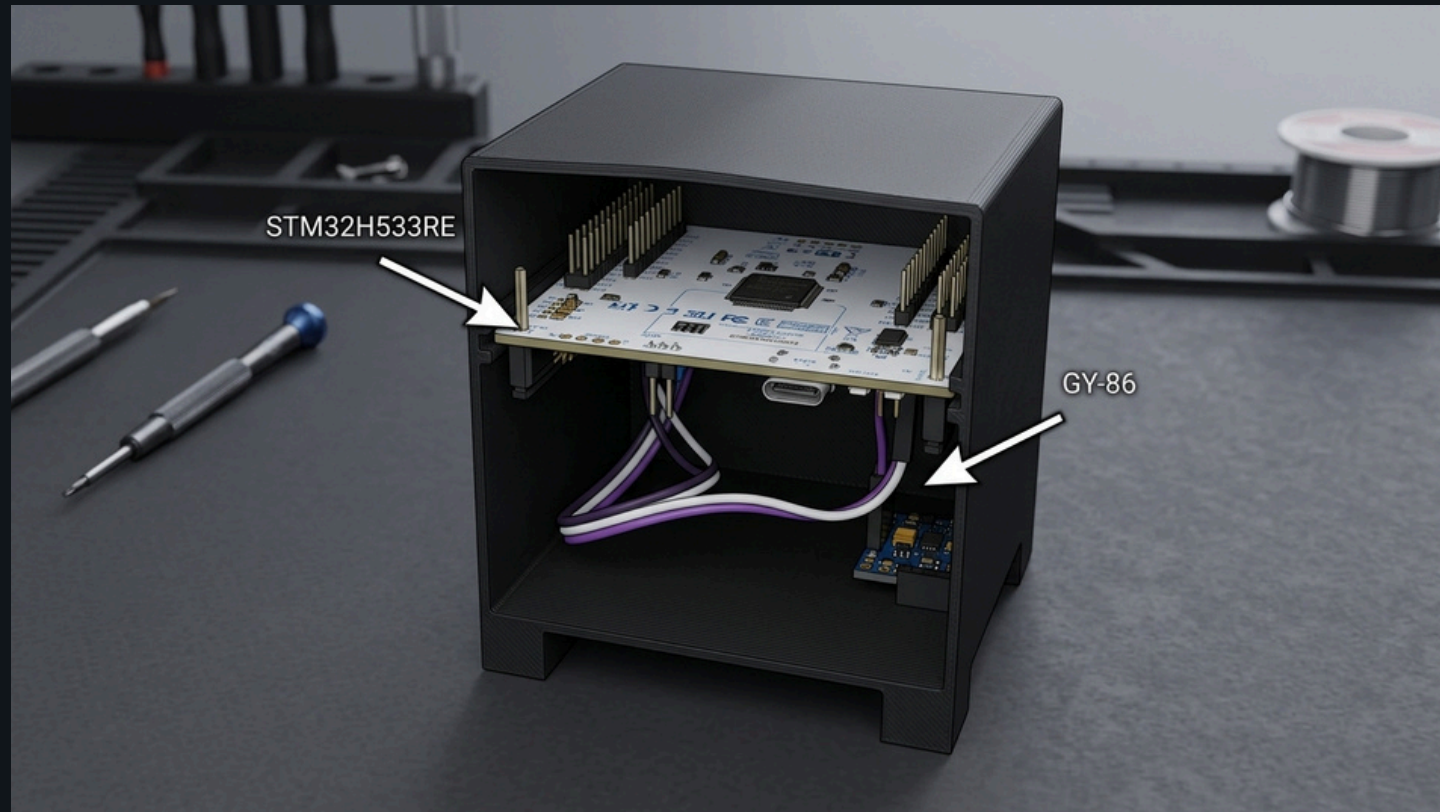
IMPLEMENTATION LAYER



RTOS is the coordinator: it does not perform the AI classification itself.

The sensor stream is structured before AI sees it

100 Hz acquisition + fixed windows make the signal repeatable for DSP and inference.



HARDWARE

STM32H533RE + GY-86

100 Hz

Sampling rate

256 samples

Frame length

2.56 s

Frame duration

50%

Overlap

1.28 s

New result interval

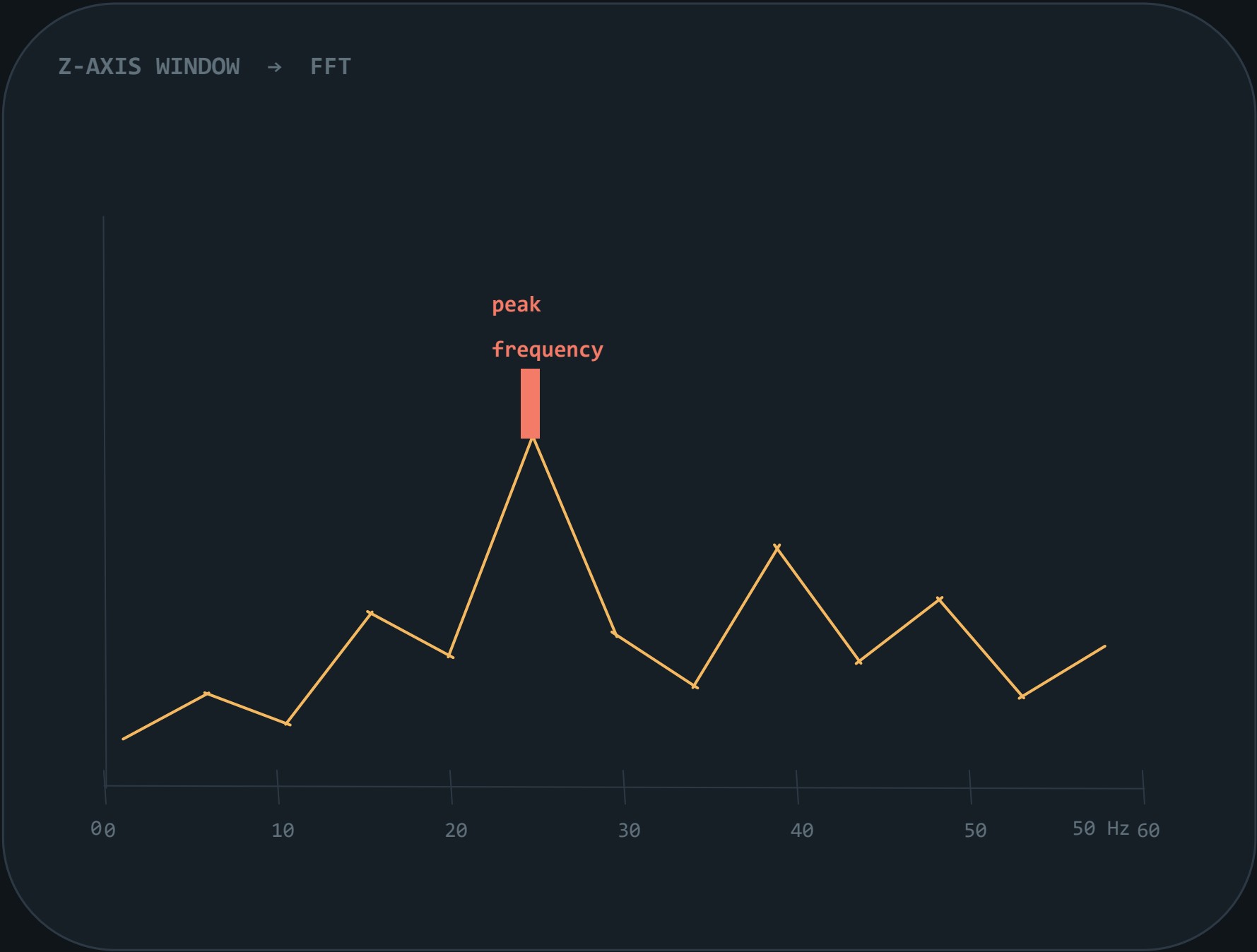
0-50 Hz

Analyzed spectrum

Z-axis acceleration is transformed into a fixed-length window for frequency analysis.

Vibration becomes six compact features

The AI does not consume the raw 256-sample waveform directly; DSP extracts a compact representation.

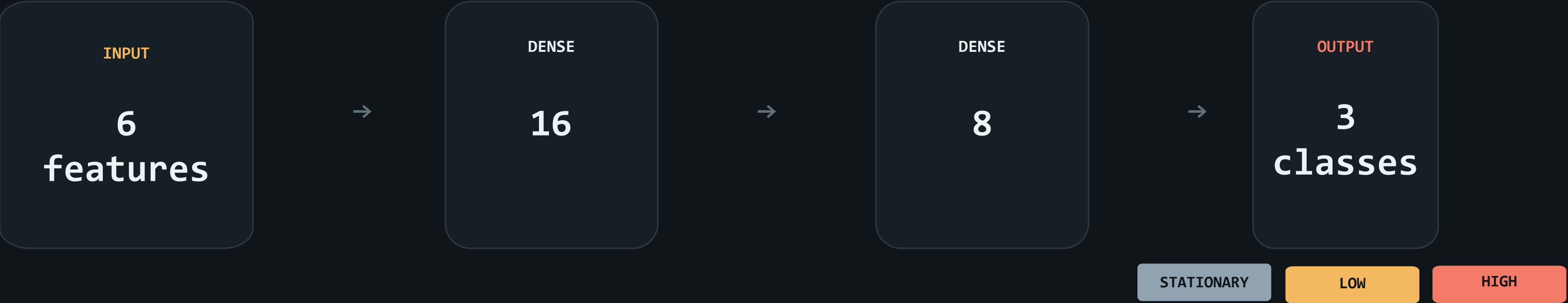


FEATURE VECTOR

01	Peak frequency	peak_hz
02	Peak amplitude	peak_mg
03	RMS	rms_mg
04	Band energy 0-5 Hz	0-5 Hz
05	Band energy 5-15 Hz	5-15 Hz
06	Band energy 15-30 Hz	15-30 Hz

A small neural network fits the edge

The deployed model is quantized and lightweight enough for embedded inference.



A_6_16_8_3_int8

TFLite / INT8

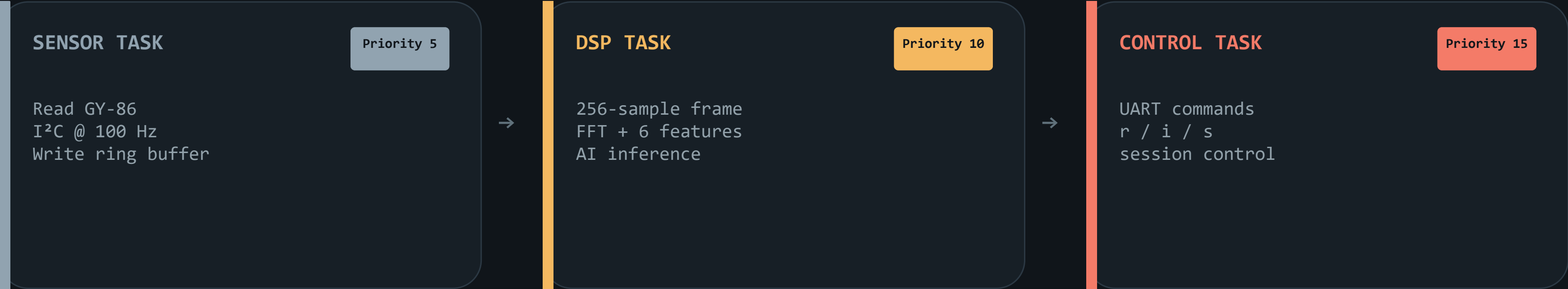
344 MACC	~1.1 KB weights	96 B activations	~10 KB AI Flash	~2 KB AI RAM
--------------------	---------------------------	----------------------------	---------------------------	------------------------

μT-Kernel 3.0 is the coordinator

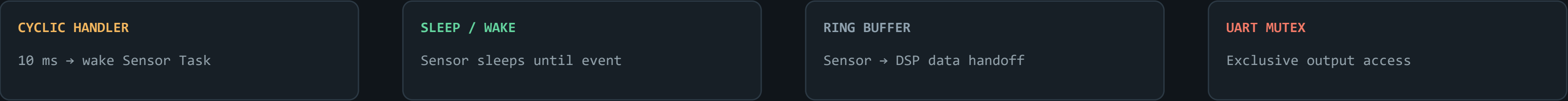
It separates periodic sensing, compute-heavy DSP/AI, and user control into concurrent firmware tasks.

μT-KERNEL 3.0

Scheduling • timing • sleep/wakeup • synchronization



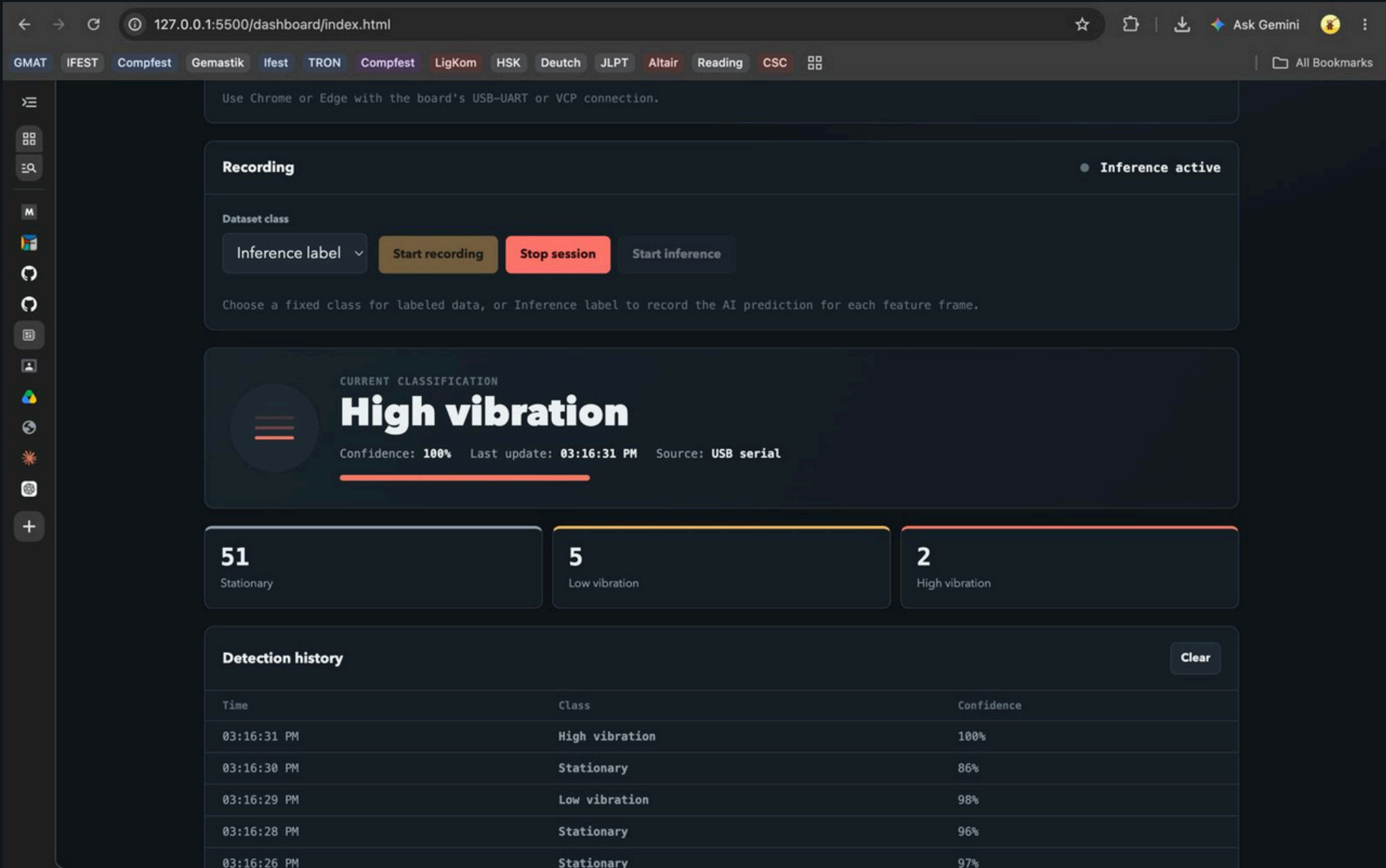
HOW THE RTOS CONNECTS THE TASKS



Priority rule: lower number = higher priority.

The dashboard makes the embedded decision observable

USB Serial connects the MCU to a browser interface for recording and inference.



CURRENT CLASSIFICATION

Class + confidence + update time.

DETECTION HISTORY

Timestamped class and confidence.

CLASS COUNTERS

Stationary / low / high totals.

RECORDING

Labeled sessions + CSV download.

Separate what is measured from what still needs testing

The repository documents runtime characteristics; dataset-level accuracy is not established here.

DOCUMENTED RUNTIME

100 Hz	sensor sampling
1.28 s	new inference opportunity
344	MACC
~1.1 KB	model weights
~10 KB	AI Flash
~2 KB	AI RAM

EXPERIMENTAL RESULT

Accuracy / Precision / Recall / F1

```
Baseline validation accuracy: 1.0000
Baseline validation macro F1: 1.0000
Baseline test accuracy: 0.9667
Baseline test macro F1: 0.9662
```

```
Validation confusion matrix:
[[60  0  0]
 [ 0 60  0]
 [ 0  0 60]]
```

```
Test confusion matrix:
[[60  0  0]
 [ 2 54  4]
 [ 0  0 60]]
```

VibeFlow closes the loop from vibration to decision

A compact embedded system combines sensing, DSP, AI, RTOS, and visualization.

SENSE

GY-86 @ 100 Hz

Capture a structured vibration signal.

INTERPRET

FFT + 6 features + INT8 AI

Convert signal characteristics into a 3-class decision.

OBSERVE

USB Serial + Dashboard

Expose confidence, history, counters, and recordings.

THE SYSTEM IN ONE LINE

Vibration → **acceleration** → **frequency features** → **Edge AI** → **class + confidence**

Next step: validate with a controlled dataset and document class-wise performance.